

Data Engineering Pipeline



*Pipeline of Data Engineering for
the Venus Brand*

What

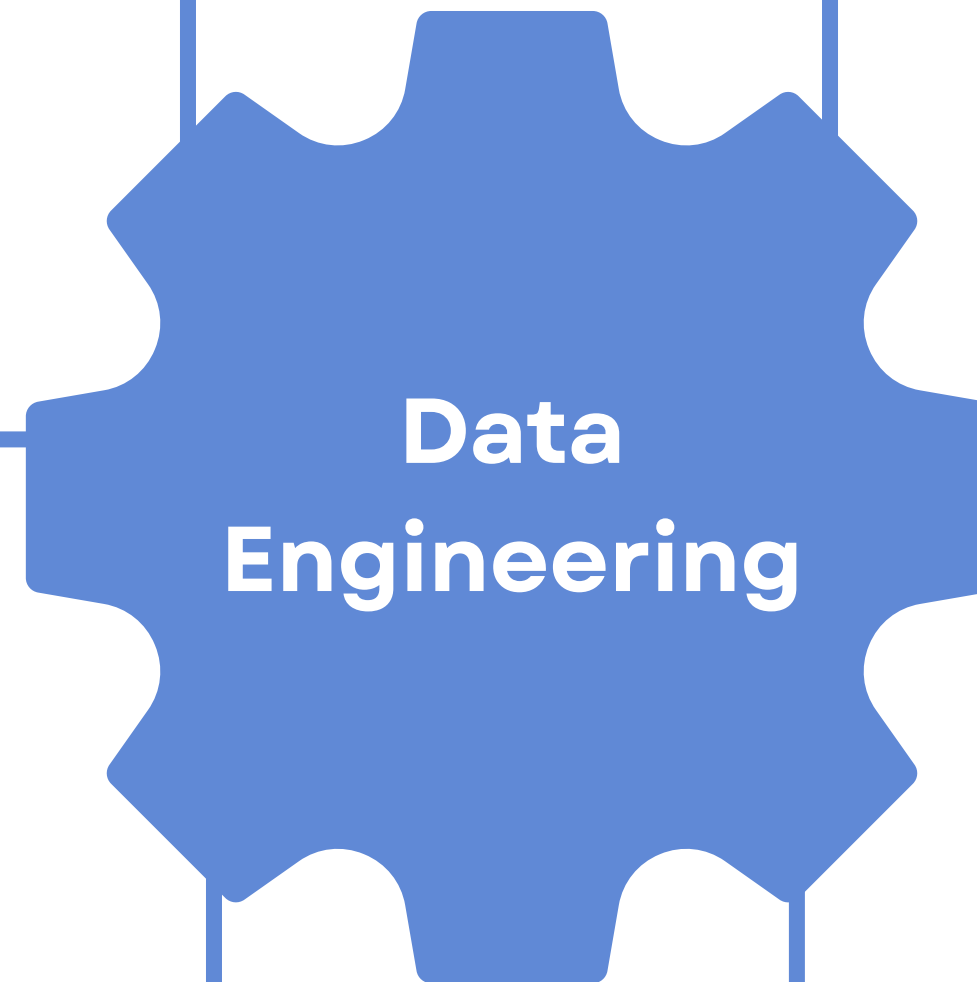
Data Engineering is how raw business data becomes trustworthy information.

When

Scheduled and depends on project type

How

- 1. Collect data from business operations*
- 2. Clean & standardize it once*
- 3. Combine it into trusted core tables*
- 4. Prepare executive-ready data for BI tools*



Why

Because leadership decisions depend on confidence in numbers.

Where

At the center of the data ecosystem.

Who

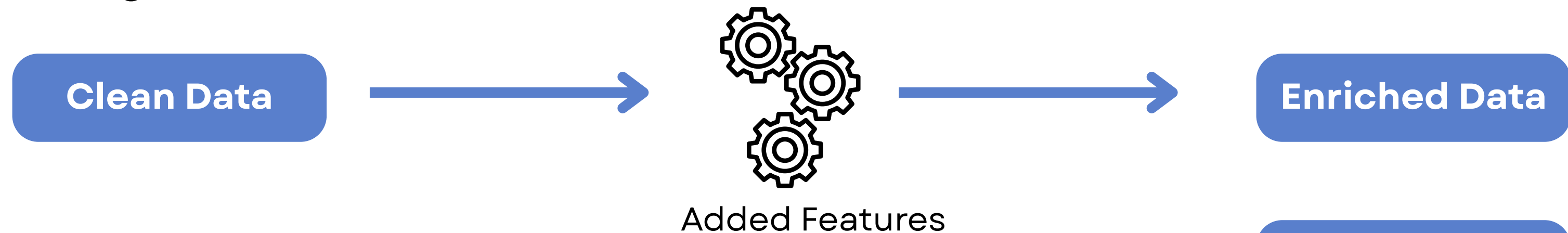
*Data engineers refines the data
Data analyst make meaningful insight with the data
Leaders asks the right question*

In short, Data Engineering Is...

Cleaning the Data



Enriching the Data



Storing and Maintaining Data



So, what's the difference?

The User must know the task in order to ask the question to the right person



Data Engineers



Data Accuracy &
Consistency

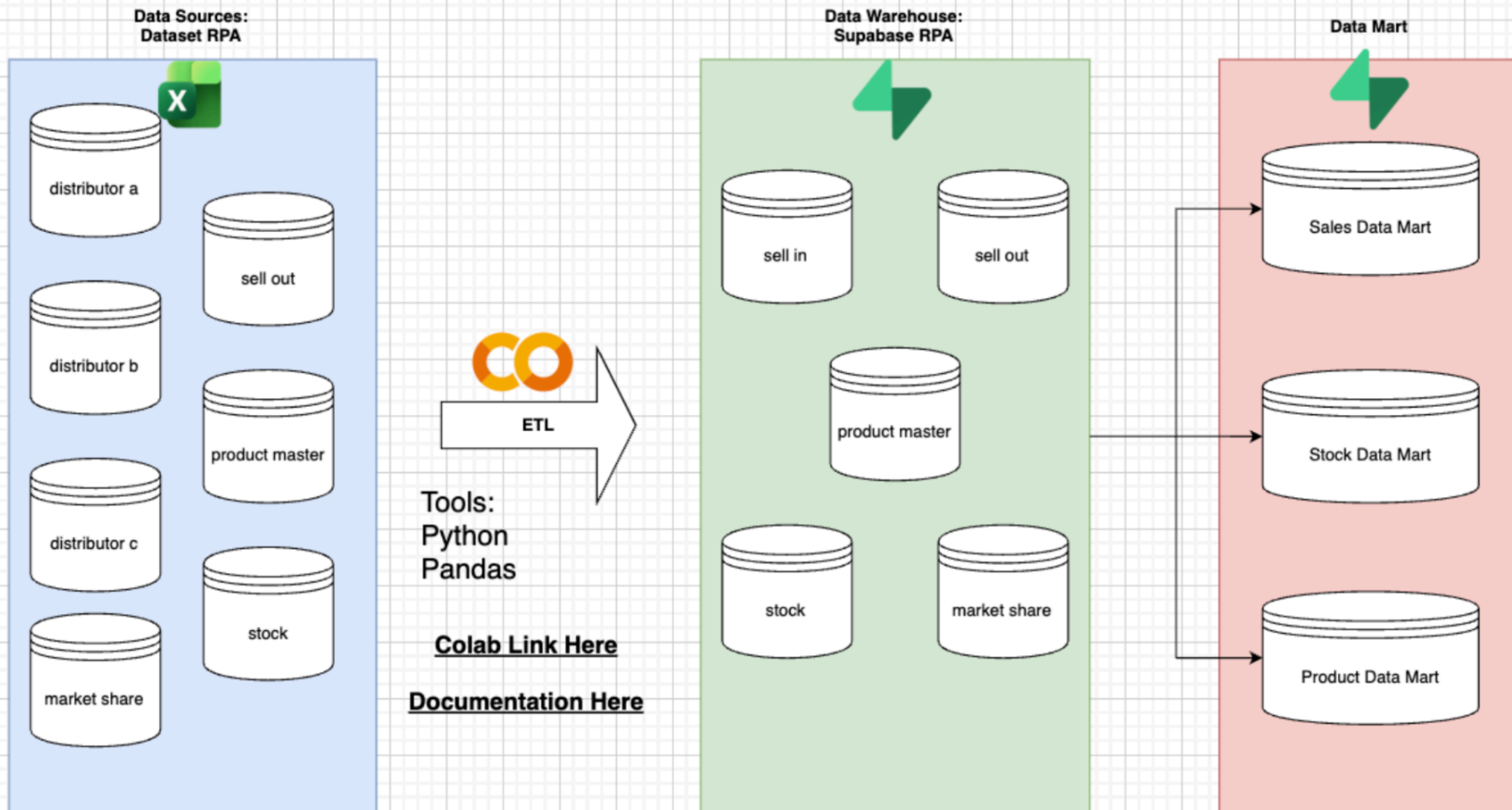


Data Analyst



Data Insights and
Patterns

Data Pipeline of RPA



Why Make Pipelines?

What This Data Pipeline Changes

What it enables

- Faster executive reporting
- One consistent set of numbers
- Clear visibility across sales, stock, and market position

Why it matters

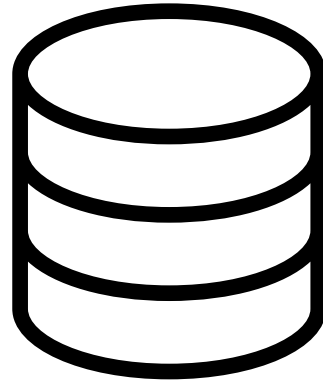
- Decisions are made with confidence
- Less time questioning data
- More time acting on insights

This pipeline reduces decision risk and increases decision speed.

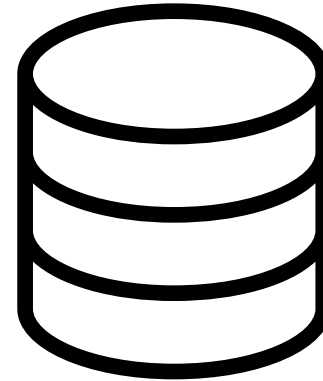


Source of Data

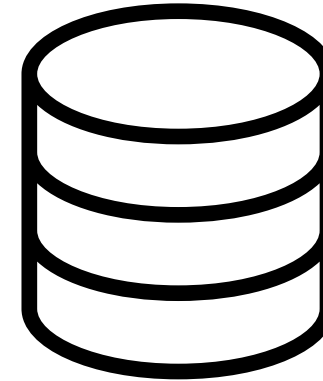
Data is provided by the client in a multi-sheet excel file



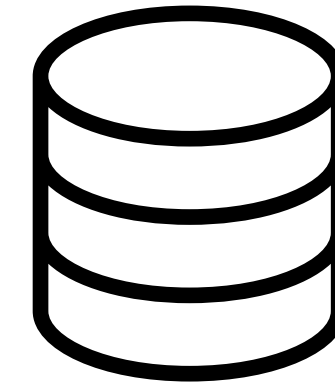
product_master



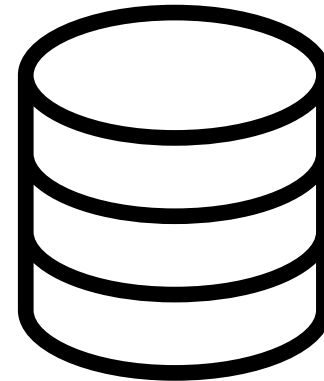
sell_out



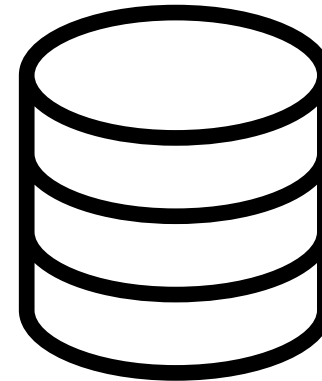
stock



market_share



sell_in_distributor_a



sell_in_distributor_b



sell_in_distributor_c

Data time frame is between July-2025 to September-2025

Sourced from 10 different stores with 25 different SKUs across 3 distributors

Clean Ups and Enrichments

What we do

- Standardize names and formats
- Remove duplicates
- Fix values that don't reflect business reality

What we add

- Product and brand context
- Master table for sell_in

Why it matters

- Prevents misleading KPIs
- Builds trust in dashboards
- Ensures leadership sees one reliable version of the truth

You may check the [pipeline documentation](#) for more detailed informations

week_da	brand	format	share
2025-07-07	brand:Venus	FACIAL WASH	0.29017614
2025-07-07	brand:Mars	FACIAL WASH	0.24847086
2025-07-07	brand:Jupiter	FACIAL WASH	0.231506492
2025-07-07	brand:Saturnus	FACIAL WASH	0.109679207
2025-07-07	brand:Uranus	FACIAL WASH	0.1201673
2025-07-07	brand:Venus	LIQUID SOAP	0.055064084
2025-07-07	brand:Mars	LIQUID SOAP	0.231958874
2025-07-07	brand:Jupiter	LIQUID SOAP	0.258602638
2025-07-07	brand:Saturnus	LIQUID SOAP	0.279143741
2025-07-07	brand:Uranus	LIQUID SOAP	0.175230663
2025-07-07	brand:Venus	MOISTURIZER	0.082336486

prefix on market_share

SERUM	BRIGHTEN	25 ML	SKU21	Venus	120490
SERUM	ENERGIZI	25 ML	SKU22	Venus	120490
SERUM	ENERGIZI	25 ML	SKU22	Venus	125490
SUNSCRE	ACNE SUP	30 ML	SKU23	Venus	39700

duplicate on product_master

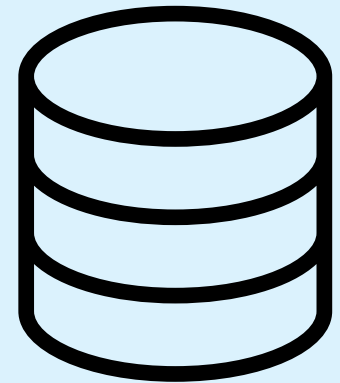
beginning_stock	stock_out	stock_in	end_stock
3	4	0	-1
3	4	0	-1

negative stock on stock

Storing the Results

The cleaned and enriched data are then stored for use in a data warehouse.

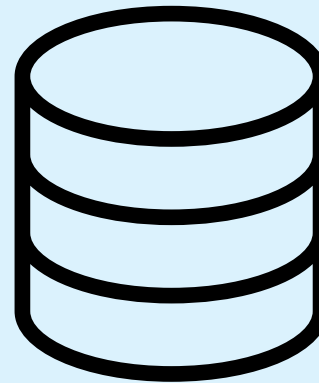
Data After Cleaning and Enriching



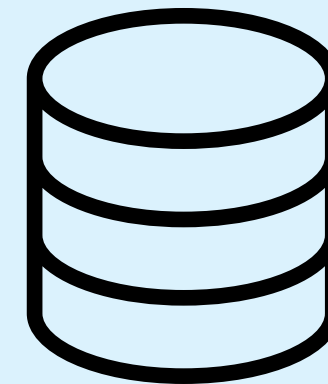
product_master



sell_out



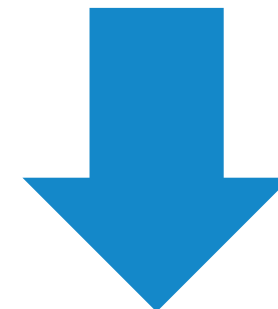
stock



market_share



sell_in



Store Data

Data Warehouse

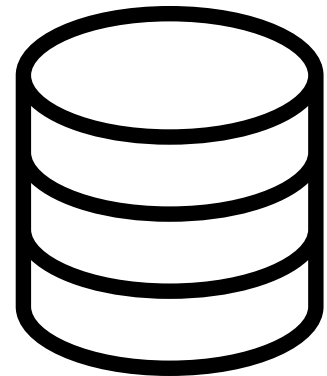


A data warehouse is a **central place** where all **cleaned** and **trusted** business **data** is stored so **leaders** can see one **consistent** version of the **truth**.

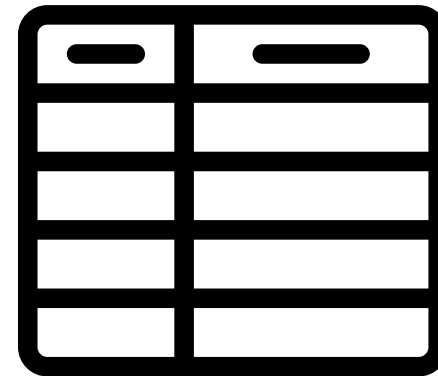
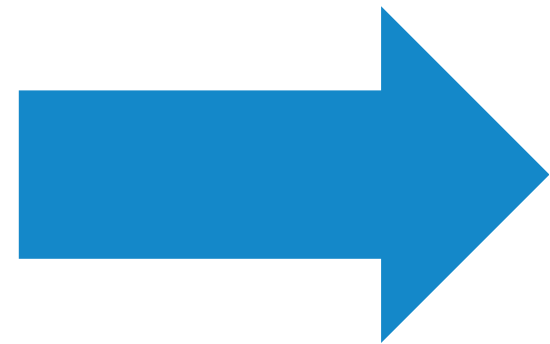
We use **Supabase (PostgreSQL)** as our data warehouse to securely store this data and power dashboards and executive reporting

Connecting the Business with Data Marts

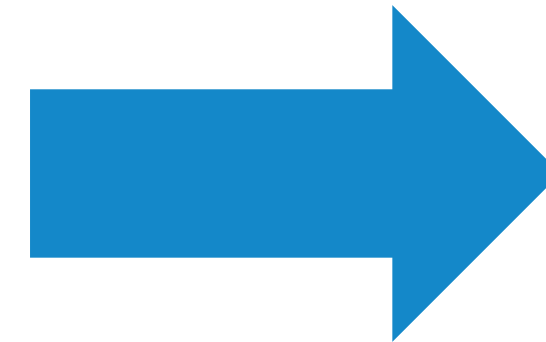
Data Marts are processed views of data referring to a certain metric, business process, or business questions.



Data Warehouse



Data Mart



Dashboard

Our Data Marts:

Stock

Are We Prepared?

Sales

Are We Growing?

Market Share

Are We Winning?

Sales Data Mart

How We Track Revenue Performance

What it shows:

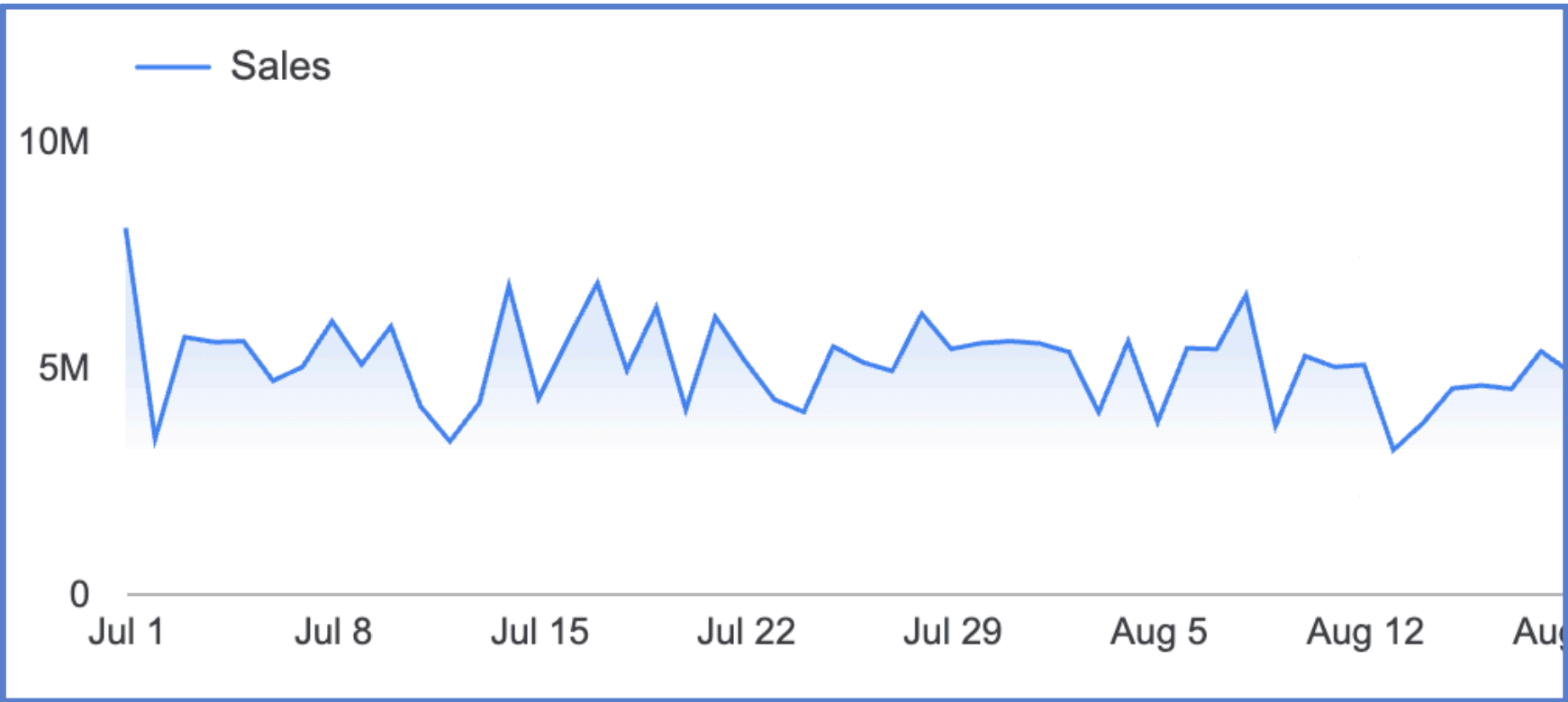
- Sales by product, store, and time
- Daily → weekly → monthly trends
- Revenue and volume in one view

value	int8	week_date	date	iso_w...	num...	mo...	num...	month_name	text	day	nu
120490		2025-06-30		27		7		July		1	
14700		2025-06-30		27		7		July		1	
77025		2025-06-30		27		7		July		1	
28770		2025-06-30		27		7		July		1	
100500		2025-06-30		27		7		July		1	
60300		2025-06-30		27		7		July		1	
120490		2025-06-30		27		7		July		1	
120490		2025-06-30		27		7		July		1	
20100		2025-06-30		27		7		July		1	
245630		2025-06-30		27		7		July		1	

Sales Data Mart

Why it matters

- Quickly see what’s growing or declining
- Compare performance consistently
- Support pricing and promotion decisions



Visualization Example

Stock Data Mart

How We Control Inventory Health

What it shows:

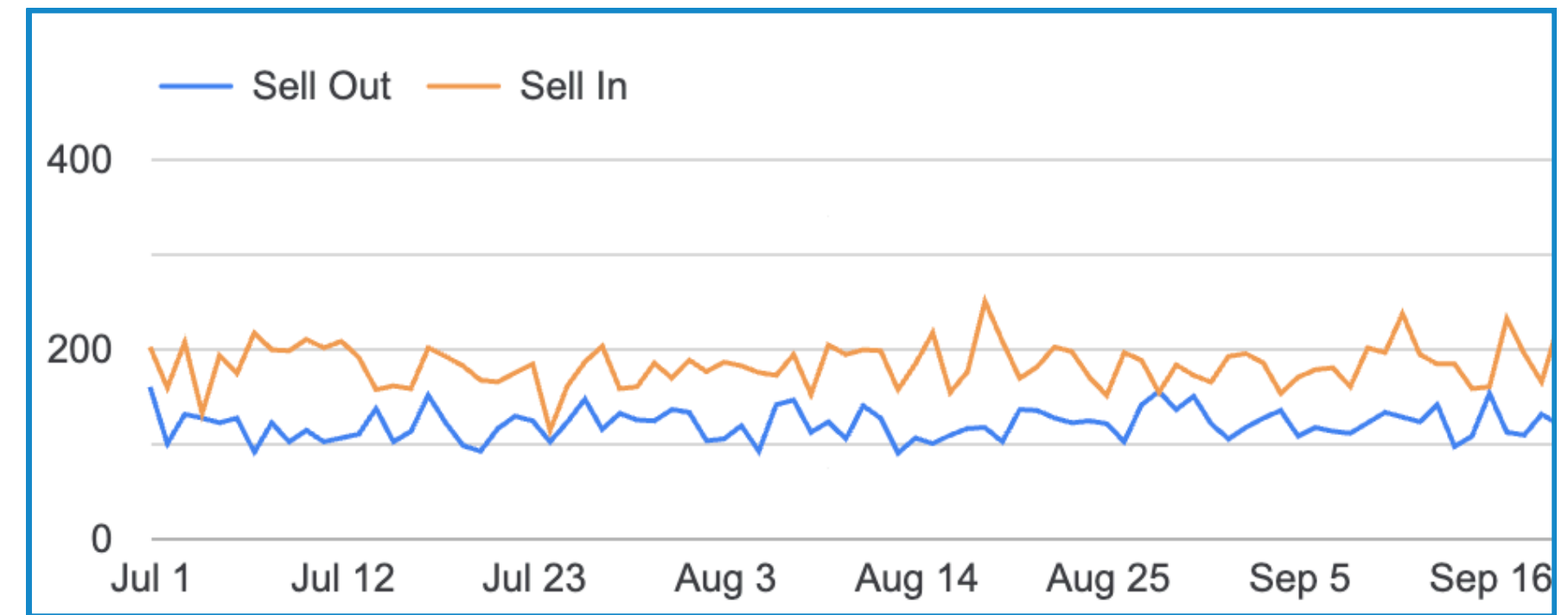
- Beginning vs ending stock
- Products sold
- Value of inventory moved

beginning_stock int4	stock_... i...	stock_in int4	end_st... i...	id int2	iso_w... num...	mo... num...
5	0	0	5	7550	27	7
5	0	4	9	9018	27	7
12	0	0	12	5797	27	7
17	0	3	20	7456	27	7
17	0	0	17	8189	27	7
10	0	0	10	8926	27	7
16	0	2	18	4880	27	7
5	0	0	5	5714	27	7
8	0	0	8	6627	27	7
16	0	3	19	7363	27	7

Stock Data Mart

Why it matters

- Detect stock-out risk early
- Reduce overstock and waste
- Improve supply planning



Visualization Example

Market Share Data Mart

How We Measure Competitive Position

What it shows

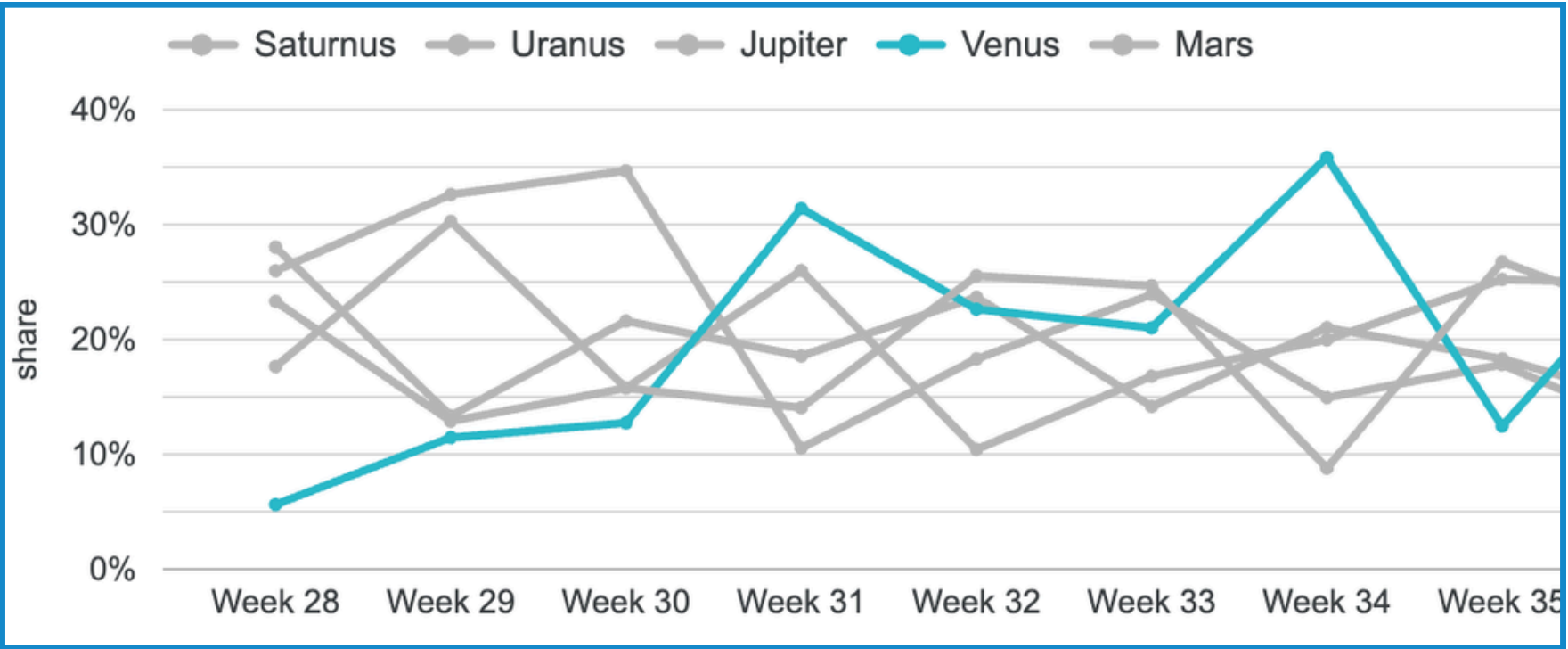
- Brand market share over time
- Weekly performance trends
- Brand-to-brand comparison

Why it matters

- Understand competitive movement
- Spot gains or losses early
- Guide brand and marketing strategy

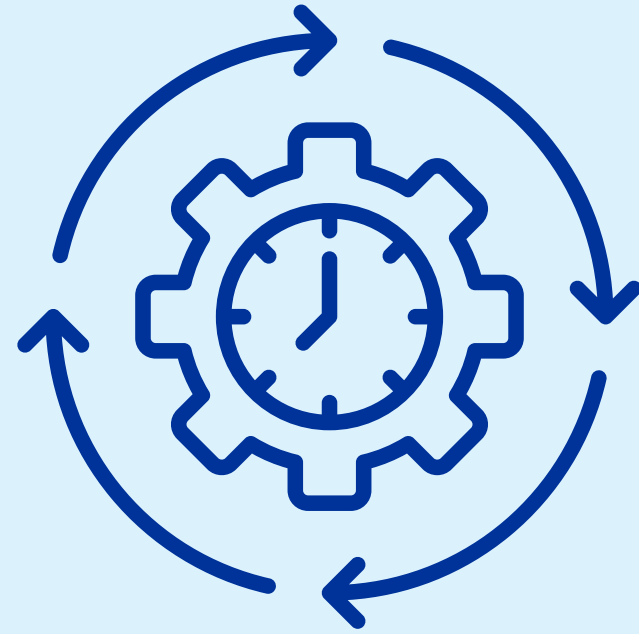
week_date	date	brand	text	format	text	share	float8
2025-07-07		Jupiter		SERUM		0.2097660347870	
25-07-07		Uranus		TONER		0.28943941970130	
2025-07-07		Saturnus		FACIAL SCRUB		0.2060786653084	
2025-07-07		Jupiter		LIQUID SOAP		0.2586026380767	
2025-07-07		Venus		SUNSCREEN		0.2058685532185	
2025-07-07		Saturnus		SUNSCREEN		0.2909880959916	
2025-07-07		Saturnus		TONER		0.3290358577256	
2025-07-07		Saturnus		MOISTURIZER		0.30206735195512	
2025-07-07		Mars		FACIAL SCRUB		0.2125400236998	

Market Share Data Mart

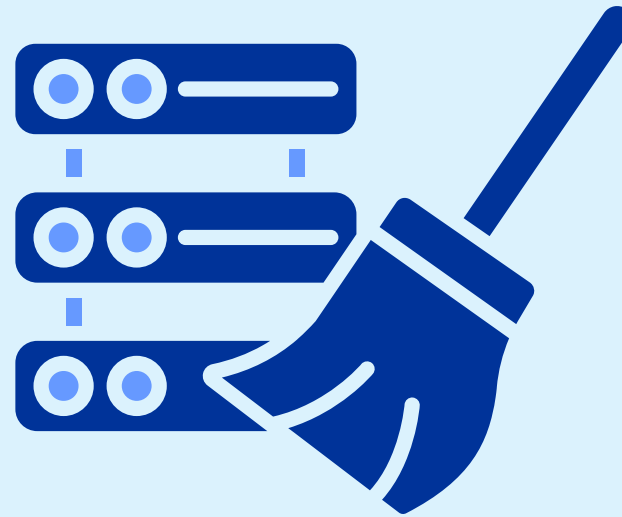


Visualization Example

What We Have Now



Data Consistency
*Allow for Repeatable
Visualization on New
Data*



Data Cleanliness
*Prevents Error on
Analysis*



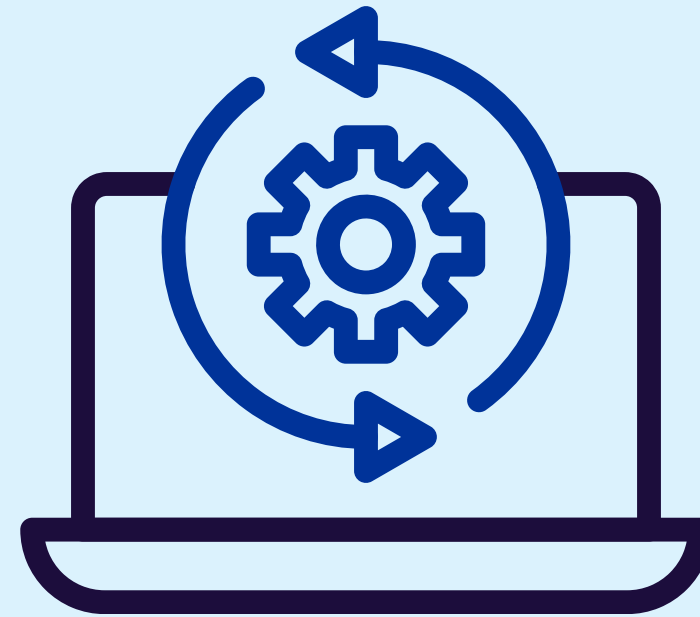
Reliable Data
*Data is Formatted
Correctly and
Logically*

Possible Improvements



Automated Data Workflow

Integrate Data Engineering Tools to Company Pipeline



More Robust Source of Data

Check on Negative Stock



Automated Alerts for Anomaly

Automated Alert for Prepared Action

Thank You
RPA SCADS DE TEAM